

Scheduling a Hospital's Operating Week

From waiting list to a feasible, urgency-aware weekly plan
Interval-based Constraint Programming · OR-Tools CP-SAT

Operations Research Scientist — Case Study

June 28, 2026



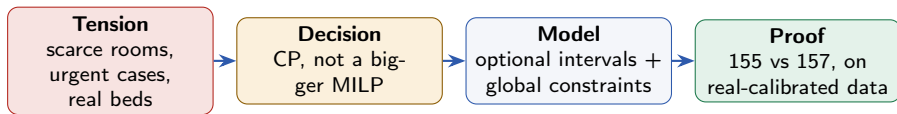
Executive walkthrough

+



Full formulation appendix

The story in one slide



What I want you to take away

- I optimise **the right thing** — clinical urgency, not raw throughput.
- I chose CP **from the problem's structure**, and I *proved* the choice.
- I know **exactly where my model is wrong**, and how I'd fix it.

Talk structure (~15 min)

1. The tension
2. The CP decision
3. The model as choices
4. Proof + reality
5. Honesty + KPIs
6. Q&A + roadmap

Full math in the appendix — sets, parameters, variables, objective, C1–C11.

+ The \$4,200-a-minute room

The setup. A hospital group's *biggest bottleneck*: deciding which elective case runs in which room, when, with which surgeon — for a whole week, at one hospital.

Why it's hard (and why it's worth doing well):

- The OR is the hospital's **largest cost centre** and top revenue line.
- Waiting lists already carry cases **past their clinical deadline**.
- Capacity is genuinely scarce: **not every case fits this week**.

The real tension

Three forces pull against each other:

- 1 **Urgency** — do the clinically pressing cases first
- 2 **Capacity** — rooms, surgeons, equipment, beds all finite
- 3 **Fairness** — who do we defer, and is that defensible?

A good plan resolves all three at once. That is the optimisation problem.

🎯 What I chose to optimise — and what I left out

In scope — what moves the needle

Case selection · room assignment · **exact intra-day timing** · surgeon day/week limits · shared equipment (real concurrency) · **recovery/ICU beds** (multi-day) · turnover buffer · the **not-this-week** decision.

Out of scope — on purpose

Stochastic durations (*extension #1*) · nurse/anaesthetist rostering (*staff pre-allocated*) · sequence-dependent cleaning (*constant buffer*) · intra-day re-optimisation (*offline plan*) · multi-week horizon.

The discipline behind the cuts ?

Every exclusion is *stated* (FORMULATION.md §11) with a reason, and **no cut ever produces an invalid schedule** — only a less flexible one. The brief explicitly rewards *choosing* what matters over modeling everything. I'd rather defend five decisions than list fifty features.

✓ The turning point: why CP, not a bigger MILP

The one sentence that decides it

“Durations sum $\leq$ capacity” is **not the same as** “cases fit without colliding” — *once a resource is shared across rooms.*

Continuous-time MILP

big-M “A-before-B”
 $O(n^2)$ binary pairs
big-M weakens LP relax.

scales
badly



CP-SAT interval

NoOverlap /
Cumulative
0 extra binaries
Theta-tree $O(n \log n)$

prunes
for free

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- A day-sum can be **tighter than reality** (forbids two genuinely non-overlapping uses).

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- CP's NoOverlap/Cumulative reason **actual clock times**, with polynomial propagation.

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This is not solver fanboyism. With no shared resources, the MILP is fine. I pick by *structure*, then *verify*.

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One variable family does the heavy lifting

For each feasible (c, d, r) surviving the eligibility pre-filter:

Optional interval per candidate slot

$$pr_{cdr} \in \{0, 1\}, \quad iv_{cdr} = \text{OptionalInterval}(\text{start}_{cdr}, t_c^{\text{tot}}, \text{end}_{cdr}, pr_{cdr})$$

Why this is the elegant part:

- The interval is *present* (binds its room, surgeon, equipment) **iff** $pr=1$.
- **One** interval per slot is shared across *all* resource constraints it touches — no channeling needed.
- Global constraints reason over *every* candidate at once.

Pre-filter = free speed

C4–C6 (roster, ambulatory, pediatric block) drop candidates **before** variables exist. The single biggest variable-count reduction — fewer vars, smaller tree, faster proof. Dominance reduction, applied up front.

Full variable definitions: appendix slide A3.



The objective, read as a clinical policy

Minimise tardiness + non-scheduling penalty

$$\min \underbrace{\sum_{dd_c \geq 0} [dd_c + d] \text{ pr}}_{\text{on-time: prefer earlier}} + \underbrace{\sum_{dd_c < 0} [dd_c + \alpha d] \text{ pr}}_{\text{overdue: } \alpha \text{ amplifies}} + \underbrace{\sum_{p_c \neq 4} p_c w_c u_c}_{\text{leave for later: costly}}$$

Each term encodes a policy a clinician would recognise:

- Tight-deadline cases scheduled *sooner* (Term 1).
- Overdue cases — deferring is *disproportionately* expensive (α , Term 2).
- Not scheduling is a *true last resort* (Term 3 dominates by construction).

Grounded, not invented ?

The tier + max-wait + escalating-penalty shape mirrors Portugal's **SIGIC** (270/60/15/3-day tiers) and the UK NHS RTT framework. Every number is an **overridable instance parameter**. *Evidence in appendix A5.*



Constraints — and where CP earns its keep

Logical / assignment

C1 ≤ 1 per patient / week

C2 priority-4 $\rightarrow$ day 1

C3 scheduled XOR penalised

C4–6 roster · ambulatory · pediatric (pre-filter)

C9 surgeon weekly cap (linear)

Full statements with math: appendix A6–A8.

Global resource constraints — the thesis

C7 Rooms · C8 Surgeon ⌚

AddNoOverlap — exact disjunction, no big-M pairs

C10 Shared equipment ⚙️

AddCumulative — literal time concurrency, capacity κ

C11 Recovery/ICU beds 🛏️

Day-scale interval $[\text{dayof}_c, \text{dayof}_c + \text{los}_c]$ — **cannot exist** in a day-bucket MILP



The demo doesn't assert the thesis — it proves it

Verified results (reproducible, tight gap):

Solver	Obj	Sched
Greedy	163	20/20
CP-SAT	155	20/20
MILP (day-bucket)	157	20/20
Gurobi	157	20/20

Why 155 beats 157 — the whole argument, concrete

The demo has one shared C-arm (capacity 1).

- MILP reads it as “ ≤ 1 C-arm case *per day, anywhere*”.
- CP-SAT places **two** C-arm cases Tuesday — different rooms, *non-overlapping* times.

That schedule is **perfectly legal**; the day-bucket MILP **forbids it outright**. CP-SAT searches a *larger, correct* feasible region.



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At 200 cases / 60s: CP-SAT **6.3% lower** obj, **+2** cases scheduled.

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That schedule is **perfectly legal**; the day-bucket MILP **forbids it outright**. CP-SAT searches a *larger, correct* feasible region.

The MILP's gap looks *tighter* (0.31% vs 4.30%) — because a smaller feasible region is easier to close. A tight gap on a too-small space is not a better answer.

➡ Calibrated to audited real data — the differentiator

The CHLN instance **reproduces published, audited statistics by construction** (Centro Hospitalar Lisboa Norte; Marques & Captivo 2015):

Metric	My instance	Published
Neurosurgery breach	261 d	~261 d
CVA breach	142 d	~147 d
% overdue (hospital)	14%	~16%

Three graded instances: **demo** (20, auditable) · **medium** (200, scalability) · **chln** (300, real-calibrated).

Honesty as a habit ⓘ

The docstring *admits* seed 7 lands near target — the exponential sampler has variance. It says “correct calibration *plus* seed luck”, and does not hide it.

Calibrating to real numbers *and* flagging the variance is the difference between an OR scientist and a number-tuner.

! What's wrong with my own model (and the fix)

Two real bugs I found reviewing the penalty code

- **Monotonicity inversion:** at equal real slack, the *more* urgent case got *less* penalty (I scale days *before* a decreasing curve). **Fix:** multiplier *after* the curve.
- **Double-counted priority:** priority enters w_c *and* again as p_c in Term 3. **Fix:** drop the extra factor.

Over-constraints

- C8: surgeon blocked during cleaning (t^{tot} , not t^{op})
- C11: horizon edge — weekend bed load uncounted

Parameters without a hard source

- $t^{\text{clean}} = 20$; multipliers 1/4.5/**18/90**
- $\alpha = 2$ — deliberately a *policy* knob

Naming my own bugs is the demonstration — not a weakness in it.

🎯 How I'd want this graded — KPIs

A low objective value is *internal*. These are the metrics I'd put in front of the hospital:

Clinical / equity

- **Overdue cases scheduled** (% of breached cases cleared this week)
- **Breach-days resolved** (sum of overdue days removed)
- **Inter-service fairness** (no specialty systematically starved)

Operational / robustness

- **Room utilisation** (target band ~80–85%: not idle, not fragile)
- **Expected overtime** (via simulation of duration overrun)
- **Day-zero cancellation rate** (robustness proxy)
- **Solve time / gap** at planning scale

The decisive validation

Retrospective pilot: take a real past week, run the model on that data, compare *against what the hospital actually did* — on these KPIs, not on my objective.

Questions I expect — and my honest answers (1/2)

Q. *Durations are deterministic and optimistic. If 20% of cases overrun, what happens Tuesday at 3pm?*

A. My biggest weakness, and extension #1. Cheapest fix: **percentile (p75) durations** — almost free, just changes the input. Then overtime penalised in the objective. Full answer: a two-stage stochastic program (L-shaped). And I'd *validate by simulation*, not by expected objective.

Q. *You compared CP against the day-bucket MILP, not the continuous big-M. Isn't that the easy opponent?*

A. Correct, and a fair hit. My comparison proves “don't use day-bucket”. It does *not* prove “CP beats big-M” — that's the textbook result (Baptiste 2001) I didn't re-measure. The honest gap in my deliverable is the missing scalability crossover between CP and big-M.

Questions I expect — and my honest answers (2/2)

Q. How much does an urgent case weigh vs a routine one? Explain it to a clinical board.

A. Today there's *no clean number* — priority enters through two mechanisms (that's the bug on the honesty slide). After the fix it's one defensible figure: “an urgent case weighs $18\times$ a routine one per day of slack” — exactly the SIGIC tier ratio a board recognises.

Q. An emergency arrives Wednesday. Do you re-run from scratch?

A. No — that reshuffles cases already promised to patients. I'd re-optimize with **stability as a cost**: LNS seeded from the live plan, freezing most of it, penalising *distance* to the communicated schedule. A 5%-worse plan that moves 2 cases beats an optimal one that moves 30.

Where I'd take it next

Near term (highest value)

- 1 Fix the two penalty bugs (one iteration)
- 2 Percentile durations → basic robustness
- 3 Loader for the real Ghent datasets (already identified)

Medium term

- 1 Two-stage stochastic (overtime recourse)
- 2 Reactive re-scheduling (LNS + stability cost)
- 3 Lexicographic objective (drop opaque weights)

A library of models

Four solver-agnostic layers: **data abstractions** (typed instances) · **constraint patterns** (capacity-sum, NoOverlap, tiered tardiness, eligibility pre-filter) · **problem templates** (this model is one) · a **thin solver-adapter** layer (one file per backend). New problem → compose existing patterns, pick a backend, justify it from structure, verify on a small instance.

Thank you

Scheduling a Hospital's Operating Week

Interval CP-SAT · defensible decisions · known limits · real-calibrated

Full formulation follows in the appendix:

sets · parameters · decision variables · objective
· constraints C1–C11 · penalty curve · evidence



Appendix

Full formulation — the actual math

A1 Problem · A2 Sets & parameters · A3 Variables · A4 Objective · A5 Penalty & evidence · A6–A8 Constraints C1–C11 · A9 Assumptions table

A1 · Problem statement (formal)

Decision

For each elective case on the waiting list, decide **(i)** whether to schedule it this week, **(ii)** on which day, **(iii)** in which room, **(iv)** at what start time, subject to surgeon, equipment and bed capacity — minimising weighted clinical tardiness plus a non-scheduling penalty.

- **Horizon:** one work week, $D = \{1, \dots, 5\}$, single hospital.
- **Surgeon** is the binding staffing resource (nurses/anaesthetists pre-rostered to rooms).
- **One** downstream bed pool modeled (recovery/ICU), constant capacity caveat (A9).
- **Offline** weekly plan — no mid-day re-optimisation (named as the top extension).

A2 · Sets, indices & parameters

Sets / indices

$c \in C$	surgical cases
$d \in D$	days, $\{1, \dots, 5\}$
$r \in R$	operating rooms
$h \in H$	surgeons
$e \in E$	shared equipment types
$D_c \subseteq D$	feasible days ($\{1\}$ if $p_c=4$)

Key parameters

$t_c^{\text{op}}, t_c^{\text{clean}}$	op + cleaning minutes
$t_c^{\text{tot}} = t_c^{\text{op}} + t_c^{\text{clean}}$	
k_{dr}	room opening minutes
k_{hd}, k_h	surgeon daily / weekly caps
$p_c \in \{1, 2, 3, 4\}$	clinical priority
dd_c	days of slack to deadline
μ_p	priority multiplier (1/4.5/18/90)
$\alpha > 1$	overdue urgency factor (2.0)
κ_{ed}	equipment concurrent capacity
los_c, β_ρ	bed stay length, pool size

A3 · Decision variables

Per feasible (c, d, r) (post eligibility pre-filter)

$$\text{pr}_{cdr} \in \{0, 1\}, \quad \text{start}_{cdr}, \text{end}_{cdr} \in [0, k_{dr}]$$

$$\text{iv}_{cdr} = \text{NewOptionalIntervalVar}(\text{start}_{cdr}, t_c^{\text{tot}}, \text{end}_{cdr}, \text{pr}_{cdr})$$

Auxiliary

$$u_c \in \{0, 1\} \text{ (unscheduled)}, \quad \sum_{d,r} \text{pr}_{cdr} + u_c = 1 \quad \forall c : p_c \neq 4$$

$\text{dayof}_c \in \{1, \dots, 5\}$ channeled from the chosen slot (for C11 beds)

iv_{cdr} is *present* iff $\text{pr}_{cdr} = 1$ — that is what lets one interval bind every resource (room, surgeon, equipment) it touches without explicit channeling.

A4 · Objective function (full)

$$\begin{aligned}
 \min \quad & \underbrace{\sum_{c:dd_c \geq 0} \sum_{d \in D_{c,r}} [dd_c + d] pr_{cdr}}_{\text{Term 1 — on-time cases, prefer earlier days}} \\
 + \quad & \underbrace{\sum_{c:dd_c < 0} \sum_{d \in D_{c,r}} [dd_c + \alpha d] pr_{cdr}}_{\text{Term 2 — overdue cases, urgency multiplier } \alpha} + \underbrace{\sum_{c:p_c \neq 4} p_c w_c u_c}_{\text{Term 3 — non-scheduling penalty}}
 \end{aligned}$$

CP-SAT requires integer coefficients; each is rounded at build time (error < 1 at this scale). Term 3's weight w_c (next slide) is calibrated to exceed any Term-1/2 coefficient, so a case is left unscheduled *only* when no feasible slot exists.

A5 · Non-scheduling penalty w_c & the evidence

$$w_c = \text{PenaltyCurve}(dd_c \cdot \mu_{p_c}) + 1.2 \cdot \max_{c' \in C} dd_{c'}$$

PenaltyCurve breakpoints (priority-1-equiv. days):

$dd \geq 90$	50
$dd \geq 0$	800
$dd \geq -15$	1000
$dd \geq -45$	2000
$dd < -45$	$2000 + 20 dd+45 $

Evidence base

- **SIGIC** (Portugal): 4 tiers, max waits 270/60/15/3 days.
- **NHS RTT** + Canadian benchmarks: same tiered-breach shape.
- Curve shape from Marques & Captivo (2015); breakpoints a generic default.

Note: this is exactly where the monotonicity-inversion bug lives — scaling $dd_c \mu_{p_c}$ before a decreasing curve.

A6 · Constraints C1–C6 (assignment & eligibility)

C1 — one occurrence per patient per week

$$\sum_{c:\text{patient}(c)=n} \sum_{d,r} \text{pr}_{cdr} \leq 1 \quad \forall n$$

C2 — priority-4 on day 1

$$\sum_r \text{pr}_{c,1,r} = 1 \quad \forall c : p_c = 4 \quad (\text{all other slots forced to 0})$$

C3 — scheduled exactly once, or penalised

$$\sum_{d,r} \text{pr}_{cdr} + u_c = 1 \quad \forall c : p_c \neq 4$$

C4 — room-service roster (pre-filter): no candidate (c, d, r) when $a_{dr}^{\text{service}(c)} = 0$.

C5 — ambulatory-only rooms admit only day-case scopes (pre-filter).

C6 — pediatric block (pre-filter): on day $d^\dagger$, service $s^\dagger$'s rooms admit only age $\leq i^\dagger$.

A7 · Constraints C7–C9 (rooms & surgeons)

C7 — room capacity via exact non-overlap

$$\text{AddNoOverlap}(\{\text{iv}_{cdr} : d, r \text{ fixed}\}) \quad \forall d, r$$

A room runs one case at a time; cases packed without collision, not merely within a minute budget.

C8 — surgeon: exact non-overlap + daily cap

$$\text{AddNoOverlap}(\{\text{iv}_{cdr} : \text{surgeon}(c) = h, d \text{ fixed}\}) \quad \forall h, d$$

$$\sum_{c:\text{surgeon}(c)=h} \sum_r t_c^{\text{op}} \text{pr}_{cdr} \leq k_{hd} \quad \forall h, d$$

A surgeon can't be in two rooms at once *and* can't exceed a daily hour budget across non-overlapping cases.

C9 — surgeon weekly limit

$$\sum_{c:\text{surgeon}(c)=h} \sum_{d,r} t_c^{\text{op}} \text{pr}_{cdr} \leq k_h \quad \forall h$$

A8 · Constraints C10–C11 (shared resources — the thesis)

C10 — shared equipment, exact concurrency

$$\text{AddCumulative}(\{\text{iv}_{cdr} : u_{c,e} = 1, d \text{ fixed}\}, \text{demands} = 1, \text{capacity} = \kappa_{ed}) \quad \forall e, d$$

Checks literal time overlap — not how many equipment- e cases land on a day. The constraint RESULTS.md shows making a measurable objective difference (155 vs 157).

C11 — downstream recovery/ICU beds (multi-day)

$$\text{dayof}_c = d_{\text{idx}} \text{ via } \text{OnlyEnforceIf}, \text{ whenever } \text{pr}_{cdr} = 1$$

$$\text{bed}_c = \text{NewOptionalIntervalVar}(\text{dayof}_c, \text{los}_c, \text{dayof}_c + \text{los}_c, \text{is_scheduled}_c)$$

$$\text{AddCumulative}(\{\text{bed}_c : \rho(c) = \rho\}, \text{demands} = 1, \text{capacity} = \beta_\rho) \quad \forall \rho$$

A day-bucket model has *no correct way* to express a multi-day stay that *starts* on the surgery day — C11 could not be added correctly without first becoming interval-based. This is the single clearest justification for the interval representation.

A9 · Assumptions & their justification

Assumption	Justification / caveat
Deterministic durations	Standard tractable approximation (Denton 2010); stochastic = extension #1
Fixed turnover buffer (t^{clean})	Constant per case; deep-clean vs quick-reset treated alike
Surgeon is binding staff resource	Nurses/anaesthetists pre-rostered to rooms — common when surgeon calendar is the bottleneck
One case per patient / week	Conservative; some services co-operate same-day multi-procedure cases
One bed pool, constant capacity	Recovery/ICU as Cumulative; day-varying capacity flagged, not modeled
Pediatric block as worked example	Shows the model absorbs ad-hoc rules as eligibility predicates, not new variables
Single week, offline	No mid-day re-optimisation; reactive rescheduling is a distinct problem (extension)

Every assumption is a

stated choice with a reason — none is a silent simplification.

A10 · Solver configuration & reproducibility

CP-SAT (primary)

- Parallel portfolio, `num_search_workers` capped at cores
- Greedy warm start via `AddHint` on (day, room) only
- No hand-written branching — OR-Tools default portfolio
- Relative gap limit 1% (configurable)

Comparison backends

- Day-bucket MILP (CBC / Gurobi) — comparison only
- Hexaly local search — optional, license-gated, graceful fallback
- Greedy — feasibility baseline + warm start source

Reproducibility

`python main.py -instance {demo,medium,chnl} -benchmark`. Six acceptance tests verify hard-constraint properties (exact non-overlap, p4 on day 1, feasibility) — the contract any reimplementations must pass. “Optimal” = proven within the configured gap, never assumed zero.